PREP 1, LESSONS 1–2

Can this data be trusted?

Mean and uncertainty; random and systematic errors; variables and choosing a graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define the resolution of a measuring instrument. [1]

2 Describe the effect of a random error on a set of readings. [1]

3 State how the effect of random error can be reduced. [1]

Key ideas LESSONS 1–2

Show your working

4 A student times a toy car rolling down a ramp three times: **2.31 s**, **2.45 s** and **2.38 s**.

a Calculate the mean time. [2]

b Calculate the uncertainty in the mean. Use uncertainty = (largest − smallest) ÷ 2 [1]

5 A balance reads 0.2 g when nothing is on it. The student uses it to measure the mass of a ball five times.

a This causes a **random** | **systematic** error. [1]

b Explain why taking more repeats will not fix this error. [1]

6 A student investigates how the **type of surface** affects the **distance** a toy car travels.

a The dependent variable is _____ **(the type of surface / the distance travelled / the mass of the car)** [1]

b Draw a ring around the correct graph: **bar chart** | **line graph** Give the reason. [1]

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